

Question #1 of 26

Question ID: 1586356

Which of the following *least accurately* identifies a type of publicly traded real estate security?

- A)** Investment trusts.
 - B)** Direct mortgage lending.
 - C)** Operating companies.
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Question #2 of 26

Question ID: 1586357

Which of the following is *most likely* to represent a publicly traded real estate debt investment?

- A)** A real estate operating company (REOC).
 - B)** Secured bank debt collateralized by real estate.
 - C)** A mortgage real estate investment trust (Mortgage REIT).
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Question ID: 1586375

Patricia Ly, CFA, is a portfolio manager who wishes to add diversification to her portfolio through the addition of a real estate investment. Ly finds the following data for a particular industrial REIT:

Net operating income (NOI): \$710,000

Funds from operations (FFO): \$630,000

Assumed cap rate: 6%

Shares outstanding: 90,000 shares

Storage property average P/FFO multiple: 13x.

Industrial property average P/FFO multiple: 10x.

Ly decides to perform a valuation on this REIT. The value per share of this REIT using a price-to-FFO approach is closest to:

- A) \$91
 - B) \$112
 - C) \$70
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Question #4 of 26

Question ID: 1586362

Which of the following is *least likely* a difference between real estate investments and traditional asset classes like stocks and bonds?

- A) Real estate tends to be indivisible
 - B) Real estate tends to be difficult to value
 - C) Real estate tends to be homogenous
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Question #5 of 26

Question ID: 1586366

Compared to transaction-based indices used to track the performance of private real estate, appraisal-based indices are *most likely* to exhibit an apparent:

- A) time lag.
- B) higher volatility.

C) higher correlation with other asset classes.

Kent Clarkson, Tony Chekov and Peter Chanwit are investment consultants for a large public pension fund. They are partners in Clarkson, Chekov and Chanwit Consulting also known as 3CC. From previous meetings with the pension board, it has been established there will be an increase in exposure to real estate for the overall portfolio. Because of the defined benefit plan's significant size and their staff's expertise, the pension fund can invest and manage all forms of real estate investments. Partners of 3CC are to recommend a form of real estate investments, and recommend potential investments.

Expected Real Estate Market Conditions

Both residential and commercial real estate prices have fallen over the last five years. This trend is not expected to persist. It is a 'buyer's market' – the current supply exceeds the current demand and prices are lower than the intrinsic value. Although interest rates have fallen to historically low rates, the volume of real estate transactions remains low. Current average 20-year commercial mortgage rates are 3.75% and expected to stay relatively flat for at least 7 more years.

Loan underwriting standards have become more stringent and loan-to-value (LTV) ratios are expected to be lower than the earlier average rate of 80%.

The four forms of real estate under consideration as an investment choice for the pension fund are:

- Private: equity option is to buy commercial properties and manage them; debt option is to directly lend to commercial property investors.
- Public: equity option is to buy equity REITs; debt option is to buy mortgage REITs or CMOs.

The following information was collected by 3CC partners to aid their analysis. The returns and standard deviations of the four possible forms of real estate investments considered are listed in Exhibit 1. Correlations of real estate index with Treasury bill returns, US aggregate bond returns and US stock returns are listed in Exhibit 2.

Exhibit 1: Returns and Standard deviation (past 20 years)

	Returns	σ
Private Equity	9.5%	6.5%
Private Debt	5.5%	8.5%
Public Equity	11.5%	21.0%

Public Debt	6.2%	22.5%
Treasuries	3.5%	0.6%

Exhibit 2: Correlation of Real Estate Index With Other Asset Classes (past 20 years)

Real Estate Index Correlations	ρ
US Treasuries	0.35
US Aggregate Bonds	-0.05
US Stocks	0.25

The partners make the following statements:

Kent Clarkson: We should eliminate the private debt option from consideration. Returns for private debt are likely to be low since interest rates are likely to remain low and the amount of underwriting that is going to be required as a lender doesn't seem worth it.

Tony Chekov: I like the equity options better than the debt options based on Clarkson's private debt expectations.

Peter Chanwit: I prefer the private option over the public option since the pension fund staff can better actively manage the real estate projects and possibly outperform the index.

The partners have identified specific REIT managers who have consistently outperformed their indices for the public option. They have also contacted potential high creditworthy borrowers in case of private debt. For the private equity option, the partners are looking at different commercial properties. They have narrowed their choices to hotels and multi-family units.

Peter Chanwit is analyzing two specific buildings. Green Oaks Hotel and Blue Ridge Apartments are next to each other; have exactly the same number of units, same amenities; were built 10 years ago by the same construction company; and managed by the same property management company. They are currently owned by different entities that are also looking to provide the financing on the following basis.

Green Oaks Hotels		Blue Ridge Apartments	
Asking Price	\$25,000,000	Asking Price	\$25,000,000
Annual NOI End of Year 1	\$2,187,500	Annual NOI End of Year 1	\$2,125,000
LTV	75.0%	LTV	70.0%

Loan Interest Rate	4.00%		Loan Interest Rate	3.50%
Monthly Debt Service	\$113,621		Monthly Debt Service	\$101,493
Loan Term	20 Years		Loan Term	20 Years
Expected Sales Price in 10 Yrs	\$30,000,000.00		Expected Sales Price in 10 Yrs	\$30,000,000.00
Principal Owed at End of 10 Yrs	\$11,222,397		Principal Owed at End of 10 Yrs	\$11,144,755

The pension fund can buy one or both buildings provided they meet the minimum criteria of a debt service coverage ratio of at least 1.50X and a levered IRR of at least 17.5%.

The indices under consideration as the benchmark for private real estate equity investing are:

- Jackson Property Index (JPI) is an appraisal based index.
- Taft's Sales Index (TSI) is a repeat sales index.
- Lincoln Hedonic Index (LHI) is a hedonic index.

Concerns regarding the index choice were verbalized at a 3CC meeting:

Kent Clarkson: I'm worried about Lincoln Hedonic Index. This index may adjust for differences in property characteristics but I'm not sure it can be effective given that some properties may not sell more than once during the index's coverage period.

Tony Chekov: I don't like the Jackson Property Index. Appraisals are estimates; there haven't been many transactions lately so I question the reliability of the returns.

Peter Chanwit: I'm not sure about Taft's Sales Index. It relies on actual transactions but there are so few sales recently so how reliable are the returns?

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Question ID: 1586351

Based on projected real estate conditions and the partners' discussion given in the vignette, 3CC's top recommendation would *most likely* be:

- A)** public debt.
 - B)** private equity.
 - C)** public equity.
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Question ID: 1586352

If the pension fund chooses to invest in hotels over apartments, one possible reason for this is that hotels:

- A)** may offer higher rates of returns because of higher operational risk.
 - B)** are commercial properties while apartments are residential properties.
 - C)** are not affected by cost and availability of debt capital.
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Question #8 - 8 of 26

Question ID: 1586353

Which statement regarding issues with indices is *least likely* correct?

- A)** Chekov's statement.
 - B)** Chanwit's statement.
 - C)** Clarkson's statement.
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Question #9 of 26

Question ID: 1586358

Which of the following *most accurately* identifies one of the disadvantages of investing in real estate through publicly traded securities? Compared to other real estate investment vehicles, publicly traded securities expose investors to:

- A)** unlimited liability.
 - B)** more-volatile returns.
 - C)** inferior liquidity.
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Question ID: 1586373

A key difference between Funds From Operations (FFO) and Adjusted Funds From Operations (AFFO) is that AFFO excludes:

- A)** non-cash rent while FFO does not.

- B) depreciation while FFO does not.
 - C) deferred tax charges while FFO does not.
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Question #11 of 26

Question ID: 1586361

Which of the following *most accurately* identifies non-core real estate property types?

- A) Office and industrial.
 - B) Hotel and hospitality.
 - C) Retail and multi-family residential.
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Question #12 of 26

Question ID: 1586355

Which of the following *most accurately* identifies one of the characteristics of a private equity investment in income-producing real estate?

- A) Sensitivity to the credit market.
 - B) Homogeneity.
 - C) Passive management.
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Question #13 of 26

Question ID: 1586359

Which of the following *most accurately* identifies one of the advantages of investing in real estate through publicly traded securities?

- A) Publicly traded corporate structures cost less to maintain.
 - B) Diversification by geography and property type is facilitated.
 - C) Structural conflicts of interest are eliminated.
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Question ID: 1586367

A real estate market is characterized by frequent transactions. However, individual properties have long holding periods. Which real estate pricing index would be *least* suitable in such an environment?

- A)** Repeat sales index.
 - B)** Hedonic price index.
 - C)** Appraisal based index.
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Question #15 of 26

Question ID: 1586363

Retail sales growth is *most likely* to be a top economic factor affecting the economic value of a(n):

- A)** industrial REIT.
 - B)** residential REIT.
 - C)** health care REIT.
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Question #16 of 26

Question ID: 1586372

The net asset value approach to valuation makes sense for REITs because:

- A)** NAV equals the value that public equity investors attach to a REIT.
 - B)** there exist active private markets for real estate assets.
 - C)** the price at which a REIT trades very closely tracks NAV.
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Question #17 of 26

Question ID: 1586376

Which of the following *most accurately* describes an approach to REIT valuation?

- A)** The discounted cash flow approach typically consists of intermediate-term cash flow projections plus a terminal value based on cash flow multiples.
- B)** The P/AFFO approach avoids estimates and assumptions in its calculation.

- C) The P/FFO approach adjusts for the impact of recurring capital expenditures needed to keep properties operating smoothly.
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Question ID: 1586360

Which of the following is the *most likely* to represent an advantage of investing in publicly traded real estate securities over direct ownership of property? Publicly traded real estate securities offer:

- A) more control over investment decisions.
 - B) greater liquidity.
 - C) lower price volatility.
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Question #19 of 26

Question ID: 1586374

Which of the following is an expense normally deducted from accounting net earnings but not from FFO?

- A) Property operating expenses
 - B) Depreciation expense
 - C) Property taxes
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Question #20 of 26

Question ID: 1586371

If a REIT has assets with a current market value of \$3,000,000, liabilities with a current market value of \$2,000,000, and 100,000 shares outstanding, what is the NAVPS per share?

- A) \$50.00
 - B) \$30.00
 - C) \$10.00
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Question #21 of 26

Question ID: 1586364

Appropriate due diligence in a private real estate investment is *most likely* to:

- A)** mitigate unforeseen potential problems.
 - B)** lower existing operating costs.
 - C)** review lease and rental history.
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Question #22 of 26

Question ID: 1586368

Compared with REITs, real estate operating companies (REOCs) are *most likely* to feature higher:

- A)** yields.
 - B)** levels of income tax exemption.
 - C)** operating flexibility.
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Question #23 of 26

Question ID: 1586354

Which of the following *most accurately* identifies a private equity investment in income-producing real estate?

- A)** Investment in a real estate investment trust (REIT).
 - B)** Direct ownership of real estate properties.
 - C)** Private market mortgage lending by an insurance company.
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Question ID: 1586369

The *most likely* consequence of the high income distribution that REITs are required to make is:

- A)** dividend yields that are nearly on-par with the yields of other publicly traded equities.

B) high volatility of reported income.

C) frequent secondary equity offerings compared to other kinds of companies.

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When calculating NAVPS, a real estate company's assets and liabilities are valued at their:

A) book value.

B) liquidation value.

C) market value.

Question #26 of 26

Question ID: 1586365

Which of the following *least accurately* describes a major category of due diligence factors that should be investigated in determining the value of a property?

A) Operating expenses.

B) Structural integrity.

C) Pipeline analysis.